Chapter 7

Working with Multivariate Calculus

7.1 Partial Derivatives

In single variable calculus the derivative $f'(x)$ of a function $f(x)$ is defined as the limit as h tends to 0 of

$$\frac{f(x+h)-f(x)}{h}.$$

The corresponding idea for multivariate calculus is the partial derivative; if $f(\mathbf{x})$ is a function of an n vector $\mathbf{x}=(x_1,x_2,...x_n)$ the partial derivative of f with respect to x_i is the limit as h_i tends to 0 of

$$\frac{f(x_1,x_2,...x_i+h_i...x_n)-f(x_1,x_2,...x_i...x_n)}{h_i}.$$

The notation for the partial derivative is either $f_i(x_1,x_2,...x_i...x_n)$ or

$$\frac{\partial f(x_1,x_2,...x_i...x_n)}{\partial x_i}$$

these expressions can be written more concisely as $f_i(\mathbf{x})$ or $\dfrac{\partial f(\mathbf{x})}{\partial x_i}$. There are issues about the existence of partial derivatives, just as there are issues about the existence of derivatives; for example the function $\min(x_1,x_2)$ does not have partial derivatives at $x_1=x_2$.

With one very important exception, the chain rule, you can use the rules for differentiation (sum, product, and quotient rules) when finding partial derivatives in exactly the same way as you use the rules finding the derivative of a function of a single variable. When finding $\frac{\partial f(\mathbf{x})}{\partial x_i}$ you simply treat $x_1,x_2,...x_{i-1},x_{i+1}...x_n$ as constants. However the chain rule is more complicated because it has to cope with the situation in which several of the arguments of a function depend upon another set of variables. I now turn to the chain rule.

7.2 The Chain Rule for Partial Derivatives

7.2.1 Where the Rule Comes From

This is essential material because you will need to use the chain rule extensively, particularly in your microeconomics course. The chain rule for functions of a single variable says that if $g(x) = f(z(x))$ then

$$\frac{dg(x)}{dx} = \frac{df(z(x))}{dz}\frac{dz(x)}{dx}.$$

The reason the chain rule holds is that the essential point about a differentiable function $f(z)$ of a single variable z is that

$$f(z) \approx f(z_0) + \frac{df(z_0)}{dz}(z - z_0)$$

when z is close to z_0. Similarly

$$z(x) \approx z(x_0) + \frac{dz(x_0)}{dx}(x - x_0)$$

when x is close to x_0. Thus if $z(x_0) = z_0$

$$g(x) - g(x_0) = f(z(x)) - f(z(x_0)) \approx \frac{df(z_0)}{dz}(z(x) - z(x_0)) \approx \frac{df(z_0)}{dz}\frac{dz(x_0)}{dx}(x - x_0)$$

so

$$\frac{g(x) - g(x_0)}{(x - x_0)} \approx \frac{df(z_0)}{dz}\frac{dz(x_0)}{dx}.$$

Taking limits gives the result. The argument as I have given it cannot be made fully rigorous because of the possibility that $z(x) - z(x_0) = 0$ at some point. Putting this right requires the assumption that the derivatives are continuous.

The chain rule for functions of many variables says that if $g(\mathbf{x}) = f(\mathbf{z}(\mathbf{x}))$ where $x = (x_1, x_2, ...x_n)$ and $z(\mathbf{x})$ is the vector of functions $(z_1(\mathbf{x}), z_2(\mathbf{x})...z_m(\mathbf{x}))$

$$\frac{\partial g(\mathbf{x})}{\partial x_i} = \sum_{j=1}^{m} \frac{\partial f(\mathbf{z}(\mathbf{x}))}{\partial z_j}\frac{\partial z_j(\mathbf{x})}{\partial x_i}.$$

The rule requires that the functions $f(\mathbf{z})$ and $z_j(\mathbf{x})$ $(j = 1, 2...m)$ have continuous partial derivatives.

The reason the chain rule holds is that the essential point about a differentiable function $f(\mathbf{z})$ of a vector of variables $z = (z_1, z_2...z_m)$ is that

$$f(\mathbf{z}) \approx f(\mathbf{z}_0) + \sum_{j=1}^{m} \frac{\partial f(\mathbf{z}_0)}{\partial z_j}(z_j - z_{0j})$$

when z is close to z_0. Similarly if x_i varies, and all the other components of x are constant

$$z_j(\mathbf{x}) \approx z_j(\mathbf{x}_0) + \frac{\partial z_j(\mathbf{x}_0)}{\partial x_i}(x_i - x_{0i})$$

when x is close to x_0. Thus if $z_0 = z(\mathbf{x}_0)$

$$\begin{aligned} g(\mathbf{x}) - g(\mathbf{x}_0) &= f(\mathbf{z}(\mathbf{x})) - f(\mathbf{z}_0) \\ &\approx \sum_{j=1}^{m} \frac{\partial f(\mathbf{z}(\mathbf{x}_0))}{\partial z_j}(z_j(\mathbf{x}) - z_j(\mathbf{x}_0)) \\ &\approx \sum_{j=1}^{m} \frac{\partial f(\mathbf{z}(\mathbf{x}_0))}{\partial z_j}\frac{\partial z_j(\mathbf{x}_0)}{\partial x_i}(x_i - x_{0i}) \end{aligned}$$

so

$$\frac{g(\mathbf{x}) - g(\mathbf{x}_0)}{(x_i - x_{0i})} \approx \sum_{j=1}^{m} \frac{\partial f(\mathbf{z}(\mathbf{x}_0))}{\partial z_j}\frac{\partial z_j(\mathbf{x}_0)}{\partial x_i}$$

Taking limits gives the result.

This argument is suggestive rather than fully rigorous both because of the possibility that $u_j(v) - u_j(v_0) = 0$ at some point, and because as I argued in the previous chapter, the approximation of the graph of the function by a plane requires continuity of partial derivatives. In practice economists almost always assume continuity of partial derivatives, usually without saying so.

7.2.2 Using the Chain Rule: an Example

Suppose that $f(z_1, z_2) = z_1^{1/4} z_2^{3/4}$. Let $g(x_1, x_2) = z_1^{1/4} z_2^{3/4}$ where $z_1(x_1, x_2) = 2x_1 + 3x_2$ and $z_2(x_1, x_2) = 4x_1 + 5x_2$. Then using the chain rule, differentiating and then substituting

$$\begin{aligned} \frac{\partial g(x_1, x_2)}{\partial x_1} &= \frac{\partial f(z_1, z_2)}{\partial z_1}\frac{\partial z_1(x_1, x_2)}{\partial x_1} + \frac{\partial f(z_1, z_2)}{\partial z_2}\frac{\partial z_2(x_1, x_2)}{\partial x_1} \\ &= \frac{\partial f(z_1, z_2)}{\partial z_1}2 + \frac{\partial f(z_1, z_2)}{\partial z_2}4 \\ &= \frac{1}{4} z_1^{-3/4} z_2^{3/4} 2 + \frac{3}{4} z_1^{1/4} z_2^{-1/4} 4 \\ &= \frac{1}{2}\left(\frac{z_2}{z_1}\right)^{3/4} + 3\left(\frac{z_1}{z_2}\right)^{1/4} \\ &= \frac{1}{2}\left(\frac{(4x_1 + 5x_2)}{(2x_1 + 3x_2)}\right)^{3/4} + 3\left(\frac{(2x_1 + 3x_2)}{(4x_1 + 5x_2)}\right)^{1/4}. \end{aligned}$$

As a check on the differentiation you can also substitute to get

$$g(x_1, x_2) = (2x_1 + 3x_2)^{1/4} (4x_1 + 5x_2)^{3/4}$$

and then differentiate using the product rule to get

$$\begin{aligned} \frac{\partial g(x_1, x_2)}{\partial x_1} &= \frac{2}{4}(2x_1 + 3x_2)^{-3/4}(4x_1 + 5x_2)^{3/4} + 3(2x_1 + 3x_2)^{1/4}(4x_1 + 5x_2)^{-1/4} \\ &= \frac{1}{2}\left(\frac{(4x_1 + 5x_2)}{(2x_1 + 3x_2)}\right)^{3/4} + 3\left(\frac{(2x_1 + 3x_2)}{(4x_1 + 5x_2)}\right)^{1/4}. \end{aligned}$$

7.2.3 The Chain Rule with Overlapping Variables: an Example

This is a very important example, because many problems in consumer theory are similar. The previous example worked with functions of two variables z_1 and z_2 which are themselves functions of two other different variables x_1 and x_2. In fact the two sets of variables often overlap as in this example.

You will learn that a consumer maximizing a Cobb-Douglas utility function $x_1^{1/3}x_2^{2/3}$ subject to a budget constraint $p_1x_1 + p_2x_2 \leq m$, where p_1 and p_2 are prices, and m is the amount of money the consumer has, buys amounts of the goods

$$\begin{aligned} x_1(p_1, p_2, m) &= \frac{1}{3}\frac{m}{p_1} \\ x_2(p_1, p_2, m) &= \frac{2}{3}\frac{m}{p_2}. \end{aligned}$$

Now suppose that $m = p_1\omega_1 + p_2\omega_2$. Economically this is the case where the consumer has a given endowment (ω_1, ω_2) rather than a given amount of money m. This is the situation in general equilibrium theory. Let

$$\begin{aligned} k_1(p_1, p_2, \omega_1, \omega_2) &= x_1(p_1, p_2, m) \\ k_2(p_1, p_2, \omega_1, \omega_2) &= x_2(p_1, p_2, m) \end{aligned}$$

when $m = p_1\omega_1 + p_2\omega_2$. You can think of the variables of the original function p_1, p_2 and m being functions of the new variables p_1, p_2, ω_1 and ω_2. The functions are

$$\begin{aligned} p_1 &= p_1 \\ p_2 &= p_2 \\ m &= p_1\omega_1 + p_2\omega_2. \end{aligned}$$

Using that chain rule

$$\begin{aligned} \frac{\partial k_1(p_1, p_2, \omega_1, \omega_2)}{\partial p_1} &= \frac{\partial x_1(p_1, p_2, m)}{\partial p_1}\frac{\partial p_1}{\partial p_1} + \frac{\partial x_1(p_1, p_2, m)}{\partial p_2}\frac{\partial p_2}{\partial p_1} + \frac{\partial x_1(p_1, p_2, m)}{\partial m}\frac{\partial m}{\partial p_1} \\ &= \frac{\partial x_1(p_1, p_2, m)}{\partial p_1} + \frac{\partial x_1(p_1, p_2, m)}{\partial m}\omega_1 \end{aligned}$$

because $\dfrac{\partial p_1}{\partial p_1} = 1$, $\dfrac{\partial p_2}{\partial p_1} = 0$ and $\dfrac{\partial m}{\partial p_1} = \omega_1$. Once you are familiar with using the chain rule in this way you can omit the first line and write down at once

$$\begin{aligned} \frac{\partial k_1(p_1, p_2, \omega_1, \omega_2)}{\partial p_1} &= \frac{\partial x_1(p_1, p_2, m)}{\partial p_1} + \frac{\partial x_1(p_1, p_2, m)}{\partial m}\frac{\partial m}{\partial p_1} \\ &= \frac{\partial x_1(p_1, p_2, m)}{\partial p_1} + \frac{\partial x_1(p_1, p_2, m)}{\partial m}\omega_1. \end{aligned}$$

As $x_1(p_1, p_2, m) = \dfrac{1}{3}\dfrac{m}{p_1}$, $\dfrac{\partial x_1(p_1, p_2, m)}{\partial p_1} = -\dfrac{1}{3}\dfrac{m}{p_1^2}$ and $\dfrac{\partial x_1(p_1, p_2, m)}{\partial m} = \dfrac{1}{3p_1}$

so

$$\begin{aligned}
\frac{\partial k_1(p_1,p_2,\omega_1,\omega_2)}{\partial p_1} &= \frac{\partial x_1(p_1,p_2,m)}{\partial p_1} + \frac{\partial x_1(p_1,p_2,m)}{\partial m}\frac{\partial m}{\partial p_1} \\
&= -\frac{1}{3}\frac{m}{p_1^2} + \frac{\omega_1}{3p_1} \\
&= -\frac{1}{3}\frac{p_1\omega_1 + p_2\omega_2}{p_1^2} + \frac{\omega_1}{3p_1} \\
&= \frac{1}{3p_1^2}\left(p_1\omega_1 - p_1\omega_1 - p_2\omega_2\right) \\
&= -\frac{p_2\omega_2}{3p_1^2}.
\end{aligned}$$

As a check $m = p_1\omega_1 + p_2\omega_2$ so

$$k_1(p_1,p_2,\omega_1,\omega_2) = \frac{p_1\omega_1 + p_2\omega_2}{3p_1} = \frac{\omega_1}{3} + \frac{p_2\omega_2}{3p_1}$$

implying that

$$\frac{\partial k_1(p_1,p_2,\omega_1,\omega_2)}{\partial p_1} = -\frac{p_2\omega_2}{3p_1^2}.$$

In this case it is much easier to do the substitution before differentiating, rather than using the chain rule. The example is here to help you become familiar with the chain rule for partial derivatives, which is important in other contexts.

7.2.4 Notation for the Chain Rule

Notation gets particularly difficult when variables overlap. The simplest case is when the function $g(x_1)$ is defined as

$$g(x_1) = f(x_1, x_2(x_1))$$

where x_2 is a function of x_1. Using the chain rule to differentiate $g(x_1)$ gives

$$\frac{dg(x_1)}{dx_1} = \frac{\partial f(x_1,x_2)}{\partial x_1} + \frac{\partial f(x_1,x_2)}{\partial x_2}\frac{dx_2(x_1)}{dx_1}.$$

Because $g(x_1)$ and $x_2(x_1)$ are functions of a single variable x_1 the derivatives $\frac{dg(x_1)}{dx_1}$ and $\frac{dx_2(x_1)}{dx_1}$ are ordinary and not partial derivatives. Sometimes $\frac{\partial f(x_1,x_2)}{\partial x_1}$ is called the partial derivative of f with respect to x_1 and $\frac{\partial f(x_1,x_2)}{\partial x_1} + \frac{\partial f(x_1,x_2)}{\partial x_2}\frac{dx_2(x_1)}{dx_1}$ is called the total derivative of f with respect to x_1.

However things get more complicated when there are more variables, for example if

$$g(x_1,x_3) = f(x_1, x_2(x_1), x_3).$$

Then

$$\frac{\partial g(x_1,x_3)}{\partial x_1} = \frac{\partial f(x_1,x_2,x_3)}{\partial x_1} + \frac{\partial f(x_1,x_2,x_3)}{\partial x_2}\frac{dx_2(x_1)}{dx_1}.$$

The term $\frac{\partial g(x_1,x_3)}{\partial x_1}$ is now a partial derivative because x_3 is being held constant. Using the term total derivative in this context may be confusing.

Different people have different ways of dealing with these notational difficulties. Once you are thoroughly familiar with using the chain rule the notation stops causing confusion, but whilst you are learning you may find it helpful to work in the following way.

I defined the functions $k_1(p_1,p_2,\omega_1,\omega_2)$ in the previous section by writing that $k_1(p_1,p_2,\omega_1,\omega_2)=x_1(p_1,p_2,m)$ when $m=p_1\omega_1+p_2\omega_2$. It is tempting to consider simply the function $x_1(p_1,p_2,p_1\omega_1+p_2\omega_2)$. But it causes problems if you want to differentiate. What do you mean by $\frac{\partial x_1}{\partial p_1}$? You could find yourself writing

$$\frac{\partial x_1(p_1,p_2,p_1\omega_1+p_2\omega_2)}{\partial p_1}=\frac{\partial x_1(p_1,p_2,p_1\omega_1+p_2\omega_2)}{\partial p_1}+\frac{\partial x_1(p_1,p_2,p_1\omega_1+p_2\omega_2)}{\partial m}\omega_1$$

where on the left hand side $\frac{\partial x_1(p_1,p_2,p_1\omega_1+p_2\omega_2)}{\partial p_1}$ is in fact $\frac{\partial k_1(p_1,p_2,\omega_1,\omega_2)}{\partial p_1}$ and on the right hand side $\frac{\partial x_1(p_1,p_2,p_1\omega_1+p_2\omega_2)}{\partial p_1}$ is in fact $\frac{\partial x_1(p_1,p_2,m)}{\partial p_1}$ evaluated at $m=p_1\omega_1+p_2\omega_2$. You may also be puzzled by what

$$\frac{\partial x_1(p_1,p_2,p_1\omega_1+p_2\omega_2)}{\partial m}$$

means, since m does not appear in $x_1(p_1,p_2,p_1\omega_1+p_2\omega_2)$.

You may find it helpful to follow these rules:

- Use different notation for different functions. Here $k_1(p_1,p_2,\omega_1,\omega_2)$ and $x_1(p_1,p_2,m)$ both give the quantity of good 1, but they give the quantity as a function of different things.
- Write out all the arguments of functions, writing for example $x_1(p_1,p_2,m)$ rather than x_1. You can use vector notation, for example writing $\mathbf{p}$ for p_1,p_2.
- When you define a new function do it by writing that $k_1(p_1,p_2,\omega_1,\omega_2)=x_1(p_1,p_2,m)$ when $m=p_1\omega_1+p_2\omega_2$ rather than writing $k_1(p_1,p_2,\omega_1,\omega_2)=x_1(p_1,p_2,p_1\omega_1+p_2\omega_2)$.

Of course following these rules makes writing things out a lot slower, and puts more on the page, which makes it visually harder to follow what is happening. Once you are thoroughly familiar with handling partial derivatives you can follow the rules in your head without writing everything down. But if an argument made with partial derivatives puzzles you it can be very helpful to write out the argument using these rules.

7.3 Directional Derivatives

Directional derivatives are an application of the chain rule. Consider the differentiable function $f:\mathcal{R}^n\to\mathcal{R}$, think of $\mathbf{x}_0$ and $\mathbf{x}_1$ as fixed n vectors and t as a number that varies and let

$$x_i = x_{0i} + t\left(x_{1i} - x_{0i}\right).$$

or in vector notation

$$\mathbf{x} = \mathbf{x}_0 + t\left(\mathbf{x}_1 - \mathbf{x}_0\right).$$

Note that this implies that

$$\frac{\partial x_i}{\partial t} = \left(x_{1i} - x_{0i}\right). \tag{7.1}$$

Let

$$g\left(t, \mathbf{x}_0, \mathbf{x}_1\right) = f\left(\mathbf{x}_0 + t\left(\mathbf{x}_1 - \mathbf{x}_0\right)\right). \tag{7.2}$$

The using the chain rule and equation 7.1

$$\begin{aligned}\frac{\partial g\left(t, \mathbf{x}_0, \mathbf{x}_1\right)}{\partial t} &= \sum_{i=1}^{n} \frac{\partial f\left(\mathbf{x}_0 + t\left(\mathbf{x}_1 - \mathbf{x}_0\right)\right)}{\partial x_i} \frac{\partial x_i}{\partial t} \\ &= \sum_{i=1}^{n} \frac{\partial f\left(\mathbf{x}_0 + t\left(\mathbf{x}_1 - \mathbf{x}_0\right)\right)}{\partial x_i}\left(x_{1i} - x_{0i}\right).\end{aligned}$$

In particular at $t = 0$ this becomes

$$\frac{\partial g\left(0, \mathbf{x}_0, \mathbf{x}_1\right)}{\partial t} = \sum_{i=1}^{n} \frac{\partial f\left(\mathbf{x}_0\right)}{\partial x_i}\left(x_{1i} - x_{0i}\right). \tag{7.3}$$

This is the directional derivative, it represents the change in the function when you move from $\mathbf{x}_0$ along the line $\mathbf{x}_0 + t\left(\mathbf{x}_1 - \mathbf{x_0}\right)$ holding $\mathbf{x}_0$ and $\mathbf{x}_1$ constant and varying.

7.4 Local Maxima and Minima

The most basic result on maximization and partial derivatives is.

Proposition 43 *Suppose that $f : \mathcal{R}^{++} \to \mathcal{R}$ is differentiable, with continuous partial derivatives, and $\mathbf{x_0}$ is a local maximum or minimum of $f(\mathbf{x})$, that is $f(\mathbf{x}) \leq f(\mathbf{x}_0)$ when $\mathbf{x}$ is close to $\mathbf{x}_0$. Then*

$$\frac{\partial f\left(\mathbf{x}_0\right)}{\partial x_i} = 0 \text{ for } i = 1, 2 \ldots n.$$

The reason behind this is the same as the reason behind the corresponding result for maximization of a function of a single variable. For such a function $f(x)$, if the derivative $df(x_0)/dx > 0$ the function is increased by increasing x slightly and decreased by decreasing x slightly so the function does not have a maximum or minimum at x_0. If the derivative $df(x_0)/dx < 0$ the function is decreased by increasing x slightly and increased by decreasing x slightly so the function does not have a maximum or minimum at x_0. Similarly if $\partial f(\mathbf{x}_0)/\partial x_i \neq 0$ the function can be increased and decreased by a small change in x_i.

7.5 Homogeneous Functions

7.5.1 Definition

A function $f(x_1, x_2, ...x_n)$ is homogeneous of degree k if for any $t > 0$

$$f(tx_1, tx_2, ...tx_n) = t^k f(x_1, x_2, ...x_n). \tag{7.4}$$

Some functions are homogeneous in some but not all variables. For example, you will meet the expenditure function $e(p_1, p_2...p_n, u)$ in consumer theory. The expenditure function $e(p_1, p_2...p_n, u)$ is a function of prices $p_1, p_2...p_n$ and utility u. The expenditure function is homogeneous of degree 1 in prices, that is

$$e(tp_1, tp_2...tp_n, u) = te(p_1, p_2...p_n, u).$$

7.5.2 Homogeneous function and derivatives

The chain rule is very useful in proving results about homogeneous functions.

Proposition 44 *If $f(x_1, x_2, ...x_n)$ is homogeneous of degree k that is*

$$f(tx_1, tx_2, ...tx_n) = t^k f(x_1, x_2, ...x_n)$$

then $\dfrac{\partial f(x_1, x_2, ...x_n)}{\partial x_i}$ is homogeneous of degree $k-1$, that is, using notation $f_i(x_1, x_2, ...x_n)$ for $\dfrac{\partial f(x_1, x_2, ...x_n)}{\partial x_i}$

$$f_i(tx_1, tx_2, ...tx_n) = t^{k-1} f_i(x_1, x_2, ...x_n).$$

Proof. Let $z_i = tx_i$ for all i so equation 7.4 becomes

$$f(z_1, z_2, ...z_n) = t^k f(x_1, x_2, ...x_n)$$

where $z_i = tx_i$ for $i = 1, 2...n$. As this equation holds for all $(x_1, x_2, ...x_n)$ the derivatives of the two sides are equal. The derivative of the left hand side with respect to x_i is

$$\frac{\partial f(z_1, z_2, ...z_n)}{\partial z_i}\frac{\partial z_i}{\partial x_i} = \frac{\partial f(tx_1, tx_2, ...tx_n)}{\partial z_i}t.$$

The derivative of the right hand side with respect to x_i is

$$t^k \frac{f(x_1, x_2, ...x_n)}{\partial x_i}$$

Equating the two sides

$$\frac{\partial f(tx_1, tx_2, ...tx_n)}{\partial z_i}t = t^k \frac{f(x_1, x_2, ...x_n)}{\partial x_i}$$

or using notation $f_i(x_1, x_2, ...x_n)$ for $\dfrac{\partial f(x_1, x_2, ...x_n)}{\partial x_i}$

$$f_i(tx_1, tx_2, ...tx_n) = t^{k-1} f_i(x_1, x_2, ...x_n).$$

(The notation gets awkward here. It is tempting to write the left hand side as $\dfrac{\partial f(tx_1, tx_2, ...tx_n)}{\partial x_i}$ but that is not the same as $\dfrac{\partial f(tx_1, tx_2, ...tx_n)}{\partial z_i}$ which is what is wanted). ■

This is a useful result. You may for example be interested in the marginal rate of technical substitution between two inputs i and j at two points $(x_1, x_2, ...x_n)$ and $(tx_1, tx_2, ...tx_n)$. These are given by

$$\frac{f_j(x_1, x_2, ...x_n)}{f_i(x_1, x_2, ...x_n)} \quad \text{and} \quad \frac{f_j(tx_1, tx_2, ...tx_n)}{f_i(tx_1, tx_2, ...tx_n)}$$

If f is homogeneous of degree k so $f_i(x_1, x_2, ...x_n)$ is homogeneous of degree $k-1$ so

$$\frac{f_j(tx_1, tx_2, ...tx_n)}{f_i(tx_1, tx_2, ...tx_n)} = \frac{t^{k-1} f_j(x_1, x_2, ...x_n)}{t^{k-1} f_i(x_1, x_2, ...x_n)} = \frac{f_j(x_1, x_2, ...x_n)}{f_i(x_1, x_2, ...x_n)}$$

and

$$\frac{f_j(tx_1, tx_2, ...tx_n)}{f_i(tx_1, tx_2, ...tx_n)} = \frac{f_j(x_1, x_2, ...x_n)}{f_i(x_1, x_2, ...x_n)}.$$

This says that the marginal rate technical of substitution is constant along a ray from the origin in the isoquant diagram.

7.6 Constant returns to scale and Euler's Theorem

You will use the result which is called Euler's Theorem in macroeconomics. (Euler was a very important eighteenth century mathematician, so a number of results carry his name.)

Definition 45 *A production function $y = f(x_1, x_2, ...x_n)$ which gives output y as a function of inputs $(x_1, x_2, ...x_n)$ displays constant returns to scale if it is homogeneous of degree 1, that is for any positive number t*

$$f(tx_1, tx_2, ...tx_n) = tf(x_1, x_2, ...x_n).$$

The way to think about constant returns to scale is that changing all inputs by the same proportion changes output by the same proportion.

Theorem 46 *Euler's Theorem states that if $f : \mathcal{R}^{++} \to \mathcal{R}^{+}$ and $f(x_1, x_2, ...x_n)$ is homogeneous of degree one then*

$$\sum_{i=1}^{n} \frac{\partial f(x_1, x_2, ...x_n)}{\partial x_i} x_i = f(x_1, x_2, ...x_n).$$

Proof. *The proof is another example of using the chain rule to do proofs. This is an important and useful technique. The key is that you must have a relationship that holds for all values of the variables in a relevant set. In this* *and many other cases in economics this is the set of vectors whose components are all non-negative. In this case the relationship is*

$$f(tx_1, tx_2, ...tx_n) = tf(x_1, x_2, ...x_n). \tag{7.5}$$

Because this relationship holds for all values of the variables, if you find the partial derivative of both sides of the equation with respect to the same variable, the partial derivatives must be equal. Differentiating both sides by t gives

$$\frac{\partial f(tx_1, tx_2, ...tx_n)}{\partial t} = f(x_1, x_2, ...x_n). \tag{7.6}$$

To find $\frac{\partial f(tx_1, tx_2, ...tx_n)}{\partial t}$ let $z_i = tx_i$ so using the chain rule

$$\begin{aligned}\frac{\partial f(z_1, z_2, ...z_n)}{\partial t} &= \sum_{i=1}^{n} \frac{\partial f(z_1, z_2, ...z_n)}{\partial z_i}\frac{\partial z_i}{\partial t} \\ &= \sum_{i=1}^{n} \frac{\partial f(z_1, z_2, ...z_n)}{\partial z_i} x_i.\end{aligned}$$

Letting $t = 1$ so $z_i = x_i$ this becomes

$$\frac{\partial f(x_1, x_2, ...x_n)}{\partial t} = \sum_{i=1}^{n} \frac{\partial f(x_1, x_2, ...x_n)}{\partial x_i} x_i.$$

Substituting this expression in the identity 7.6 the identity becomes

$$\sum_{i=1}^{n} \frac{\partial f(x_1, x_2, ...x_n)}{\partial x_i} x_i = f(x_1, x_2, ...x_n).$$

■

The result is economically important, because if a firm is a price taker in both its output and input markets and the price of the output is p and the price of input i is w_i the first order conditions for profit maximization imply that

$$p\frac{\partial f(x_1, x_2, ...x_n)}{\partial x_i} = w_i$$

so taken with Euler's theorem this implies that

$$pf(x_1, x_2, ...x_n) = \sum_{i=1}^{n} w_i x_i$$

so the firm makes zero profits. The revenue from production is all distributed to the factors of production. This is a condition relating prices to the technology of the firm. The condition is required for an equilibrium in which the firm is a price taker. If it is not satisfied there are two possibilities. One is that the firm makes losses at any positive level of output so it shuts down. The other is that it can make strictly positive profits at some level of output, so with constant returns to scale it can expand output to make infinitely large profits so long as it is a price taker. At some stage it will have a large market share and it will no longer be a price taker.

Production functions are also used in growth theory in macroeconomics, where the function gives the output of an entire economy as a function of inputs, often capital and labour. In this context Euler's theorem says that the entire national income is distributed to the factors of production.

7.7 Second derivatives and Young's Theorem

If $f(\mathbf{x})$ is a function of $\mathbf{x} = (x_1, x_2, ...x_n)$ the second partial derivative of the $f(\mathbf{x})$ with respect to x_i is defined analogously to the second derivative of a function of a single variable so

$$\frac{\partial^2 f(\mathbf{x})}{\partial x_i^2} = \frac{\partial \left(\frac{\partial f(\mathbf{x})}{\partial x_i}\right)}{\partial x_i}.$$

Functions of many variables also have cross partial derivatives

$$\frac{\partial^2 f(\mathbf{x})}{\partial x_j \partial x_i} = \frac{\partial \left(\frac{\partial f(\mathbf{x})}{\partial x_i}\right)}{\partial x_j}.$$

It is not always true that

$$\frac{\partial^2 f(\mathbf{x})}{\partial x_j \partial x_i} = \frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j}.$$

But it is true that

$$\frac{\partial^2 f(\mathbf{x})}{\partial x_j \partial x_i} = \frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j}$$

if $f(\mathbf{x})$ has continuous second derivatives. This is Young's theorem. The matrix of second derivatives

$$\left[\frac{\partial^2 f(\mathbf{x})}{\partial x_j \partial x_i}\right]$$

is called the Hessian matrix.

7.8 The Implicit Function Theorem

Implicit functions are important and familiar in economics. You will be familiar with drawing indifference curves in consumer theory. These show x_2 as a function of x_1, but are derived from the equation $u(x_1, x_2) = u_1$ for some fixed number u_1. Different indifference curves correspond to different values u_1, u_2.... Sometimes you can derive x_2 as an explicit function of x_1 and u_1 by rearranging the equation $u(x_1, x_2) = u_1$, sometimes this is difficult, or impossible. But this does not fundamentally matter if you can be sure the function exists and can find its derivatives.

Existence can be a problem. You want to find the function $y = f(x)$ defined implicitly by the equation

$$x^2 + y^2 = 25.$$

If you solve this equation for y you get solutions

$$y = \sqrt{(25 - x^2)}$$

and

$$y = -\sqrt{(25 - x^2)}.$$

There is a unique solution $y = 0$ if $x = 5$ or $x = -5$. If $x < -5$ or $5 < x$ there are no real valued solutions and if $-5 < x < 5$ there are two solutions. By definition a function of x has a unique value for each value of x. The most that can be asked for is uniqueness in a neighbourhood. For example if (x, y) has to be is close to $(3, 4)$ the only is solution $y = \sqrt{(25 - x^2)}$. Even so there can be problems. There are two solutions in any neighbourhood of $(5, 0)$. The implicit function theorem gives conditions under which an implicit function exists, and shows how to find its derivative.

Theorem 47 (The implicit function theorem) *Suppose that* $G(x_1, x_2, ..x_k, y)$ *is a continuous function with continuous partial derivatives defined in a neighbourhood of a point* $(x_1^*, x_2^*, ..x_k^*, y^*)$, *with*

$$G(x_1^*, x_2^*, ..x_k^*, y^*) = c$$

and

$$\frac{\partial G(x_1^*, x_2^*, ..x_k^*, y^*)}{\partial y} \neq 0.$$

Then there is function $y(x_1, x_2....x_k)$ *with continuous partial derivatives defined on a neighbourhood of* $(x_1^*, x_2^*, ..x_k^*)$ *with the property that*

$$G(x_1, x_2....x_k, y(x_1, x_2....x_k)) = c$$

$$y^* = y(x_1^*, x_2^*, ..x_k^*)$$

and

$$\frac{\partial y(x_1^*, x_2^*, ..x_k^*)}{\partial x_i} = -\frac{\dfrac{\partial G(x_1^*, x_2^*, ..x_k^*, y^*)}{\partial x_i}}{\dfrac{\partial G(x_1^*, x_2^*, ..x_k^*, y^*)}{\partial y}}.$$

The difficult part of the theorem is proving existence. The result on the derivative follows easily from the chain rule. Suppose that

$$G(x_1, x_2, ..x_k, y(x_1, x_2, ..x_k)) = c \tag{7.7}$$

where c is a constant, Using the chain rule the derivative of $G(x_1, x_2, ..x_k, y(x_1, x_2, ..x_k))$ with respect to x_i is

$$\frac{\partial G(x_1, x_2, ..x_k, y)}{\partial x_i} + \frac{\partial G(x_1, x_2, ..x_k, y)}{\partial y}\frac{\partial y(x_1, x_2, ..x_k)}{\partial x_i} = 0$$

as $\dfrac{\partial c}{\partial x_i} = 0$. Thus if $\dfrac{\partial G(x_1, x_2, ..x_k, y)}{\partial y} \neq 0$

$$\frac{\partial y(x_1, x_2, ..x_k)}{\partial x_i} = -\frac{\dfrac{\partial G(x_1, x_2, ..x_k, y)}{\partial x_i}}{\dfrac{\partial G(x_1, x_2, ..x_k, y)}{\partial y}}.$$

7.9 Taylor's Expansion with Multivariate Calculus

7.9.1 Taylor's approximation with a single variable

In the chapter on Taylor series in the Background Notes I showed that a function $f(x)$ evaluated at x_0 can be approximated by a polynomial of degree m. At x_0 the value of the function and its first, second ....m^{th} derivatives are the same as the corresponding values of the polynomial. The polynomial is

$$f(x_0) + \sum_{k=1}^{m} \frac{f^k(x_0)}{k!}(x - x_0)^k.$$

This section shows how the result can be extended to functions of many variables, and in particular demonstrates that the quadratic approximation is

$$f(\mathbf{x}) \approx f(\mathbf{x}_0) + \sum_{i=1}^{n} \frac{\partial f(\mathbf{x}_0)}{\partial x_i}(x_{1i} - x_{0i}) + \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n} \frac{\partial^2 f(\mathbf{x}_0)}{\partial x_i \partial x_j}(x_i - x_{0i})(x_j - x_{0j}).$$

The right hand side of this relation is a quadratic form that can be written in vector notation as

$$f(\mathbf{x}) \approx f(\mathbf{x}_0) + \left(\frac{\partial f(\mathbf{x}_0)}{\partial \mathbf{x}}\right)'(\mathbf{x} - \mathbf{x}_0) + \frac{1}{2}(\mathbf{x} - \mathbf{x}_0)' \frac{\partial^2 f(\mathbf{x}_0)}{\partial \mathbf{x}^2}(\mathbf{x} - \mathbf{x}_0)$$

where $\frac{\partial f(\mathbf{x}_0)}{\partial \mathbf{x}}$ is an n vector whose i^{th} component is $\frac{\partial f(\mathbf{x}_0)}{\partial x_i}$ and $\frac{\partial^2 f(\mathbf{x}_0)}{\partial \mathbf{x}^2}$ is an $n \times n$ matrix for which component ij is $\frac{\partial^2 f(\mathbf{x}_0)}{\partial x_i \partial x_j}$.

7.9.2 Proving the Result

The result is proved using the same trick as I used when defining a directional derivative partial derivatives. This works with the function $g(t, \mathbf{x_0}, \mathbf{x_1})$ defined as

$$g(t, \mathbf{x_0}, \mathbf{x_1}) = f(\mathbf{x}_0 + t(\mathbf{x_1} - \mathbf{x_0}))$$

Thinking of $\mathbf{x}_0$ and $\mathbf{x}_0$ as fixed, and treating $g(t, \mathbf{x_0}, \mathbf{x_1})$ as a function of a single variable t the polynomial approximation is

$$g(t, \mathbf{x_0}, \mathbf{x_1}) \approx g(0, \mathbf{x_0}, \mathbf{x_1}) + \sum_{k=1}^{m} \frac{1}{k!} \frac{\partial^k g(0, \mathbf{x_0}, \mathbf{x_1})}{\partial t^k} t^k.$$

It is possible in principle to use the chain rule to write down the derivative

$$\frac{\partial^k g(0, \mathbf{x_0}, \mathbf{x_1})}{\partial t^k}$$

in terms of f, $\mathbf{x}_0$, $\mathbf{x}_1$ and t. In practice economists confine themselves to the quadratic case with $m = 2$,

$$g(t, \mathbf{x_0}, \mathbf{x_1}) \approx g(0, \mathbf{x_0}, \mathbf{x_1}) + t\frac{\partial g(0, \mathbf{x_0}, \mathbf{x_1})}{\partial t} + \frac{1}{2}t^2 \frac{\partial^2 g(0, \mathbf{x_0}, \mathbf{x_1})}{\partial t^2}. \tag{7.8}$$

The next task is getting an expressions for the right hand side of this relation in terms of f, $\mathbf{x}_0$, $\mathbf{x}_1$ and t. From the definition of g

$$g\left(0, \mathbf{x_0}, \mathbf{x_1}\right) = f\left(\mathbf{x}_0\right). \tag{7.9}$$

The chain rule implies that

$$\frac{\partial g\left(t, \mathbf{x}_0, \mathbf{x}_1\right)}{\partial t} = \sum_{j=1}^{n} \frac{\partial f\left(\mathbf{x}_0 + t\left(\mathbf{x_1} - \mathbf{x_0}\right)\right)}{\partial x_j}\left(x_{1j} - x_{0j}\right) \tag{7.10}$$

and in particular that

$$\frac{\partial g\left(0, \mathbf{x}_0, \mathbf{x}_1\right)}{\partial t} = \sum_{j=1}^{n} \frac{\partial f\left(\mathbf{x}_0\right)}{\partial x_j}\left(x_{1j} - x_{0j}\right). \tag{7.11}$$

I now want the second derivative $\dfrac{\partial^2 g\left(t, \mathbf{x}_0, \mathbf{x}_1\right)}{\partial t^2}$, so I need to differentiate the right hand side of equation 7.10 with respect to t. Using the chain rule for partial derivatives the derivative of

$$\frac{\partial f\left(\mathbf{x}_0 + t\left(\mathbf{x_1} - \mathbf{x_0}\right)\right)}{\partial x_j}$$

with respect to t is

$$\sum_{i=1}^{n} \frac{\partial^2 f\left(\mathbf{x}_0 + t\left(\mathbf{x_1} - \mathbf{x_0}\right)\right)}{\partial x_i \partial x_j}\left(x_{1i} - x_{0i}\right)$$

so from equation 7.11

$$\frac{\partial^2 g\left(t, \mathbf{x}_0, \mathbf{x}_1\right)}{\partial t^2} = \sum_{i=1}^{n}\sum_{j=1}^{n} \frac{\partial^2 f\left(\mathbf{x}_0 + t\left(\mathbf{x_1} - \mathbf{x_0}\right)\right)}{\partial x_i \partial x_j}\left(x_{1i} - x_{0i}\right)\left(x_{1j} - x_{0j}\right)$$

and in particular at $t = 0$

$$\frac{\partial^2 g\left(0, \mathbf{x}_0, \mathbf{x}_1\right)}{\partial t^2} = \sum_{i=1}^{n}\sum_{j=1}^{n} \frac{\partial f^2\left(\mathbf{x}_0\right)}{\partial x_i \partial x_j}\left(x_{1i} - x_{0i}\right)\left(x_{1j} - x_{0j}\right) \tag{7.12}$$

Thus from equations 7.8, 7.9, 7.11 and 7.12

$$\begin{aligned}
& f\left(\mathbf{x}_0 + t\left(\mathbf{x_1} - \mathbf{x_0}\right)\right) \\
= \quad & g\left(t, \mathbf{x}_0, \mathbf{x}_1\right) && (7.13) \\
\approx \quad & g\left(0, \mathbf{x}_0, \mathbf{x}_1\right) + t\frac{\partial g\left(0, \mathbf{x}_0, \mathbf{x}_1\right)}{\partial t} + \frac{1}{2}t^2\frac{\partial^2 g\left(0, \mathbf{x}_0, \mathbf{x}_1\right)}{\partial t^2} \\
= \quad & f\left(\mathbf{x}_0\right) + t\left(\sum_{j=1}^{n} \frac{\partial f\left(\mathbf{x}_0\right)}{\partial x_j}\left(x_{1j} - x_{0j}\right)\right) \\
& + \frac{1}{2}t^2\left(\sum_{i=1}^{n}\sum_{j=1}^{n} \frac{\partial^2 f\left(\mathbf{x}_0\right)}{\partial x_i \partial x_j}\left(x_{1i} - x_{0i}\right)\left(x_{1j} - x_{0j}\right)\right). && (7.14)
\end{aligned}$$

Now let

$$\mathbf{x} = \mathbf{x}_0 + t\left(\mathbf{x_1} - \mathbf{x_0}\right)$$

so

$$\begin{aligned} g\left(t, \mathbf{x}_0, \mathbf{x}_1\right) &= f\left(\mathbf{x}\right) \\ t\left(x_{1j} - x_{0j}\right) &= x_j - x_{0j} \\ t^2\left(x_{1i} - x_{0i}\right)\left(x_{1j} - x_{0j}\right) &= \left(x_i - x_{0i}\right)\left(x_j - x_{0j}\right) \end{aligned}$$

and substituting in equation 7.14 gives the result

$$f\left(\mathbf{x}\right) \approx f\left(\mathbf{x}_0\right) + \sum_{j=1}^{n} \frac{\partial f\left(\mathbf{x}_0\right)}{\partial x_j}\left(x_j - x_{0j}\right) + \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n} \frac{\partial f^2\left(\mathbf{x}_0\right)}{\partial x_i \partial x_j}\left(x_i - x_{0i}\right)\left(x_j - x_{0j}\right). \tag{7.15}$$

This can also be written in vector and matrix notation. Let $Df\left(\mathbf{x}_0\right)$ be a column vector whose i^{th} component is $\frac{\partial f(\mathbf{x}_0)}{\partial x_j}$. Let $\mathbf{x} - \mathbf{x}_0$ be a column vector whose i^{th} component is $x_i - x_{i0}$ and let $D^2 f\left(\mathbf{x}_0\right)$ be an $n \times n$ matrix for which component ij is $\frac{\partial f^2(\mathbf{x}_0)}{\partial x_i \partial x_j}$. Then equation 7.15 becomes

$$f\left(\mathbf{x}_0\right) \approx f\left(\mathbf{x}_0\right) + Df\left(\mathbf{x}_0\right)'\left(\mathbf{x} - \mathbf{x_0}\right) + \frac{1}{2}\left(\mathbf{x} - \mathbf{x_0}\right)' D^2 f\left(\mathbf{x}_0\right)\left(\mathbf{x} - \mathbf{x_0}\right).$$